

DATA VORTEX 2026

ROUND 4 — DASHBOARD USER GUIDE & DOCUMENTATION

Interactive Architecture, Navigation Guide, and Execution Manual

September 2026 | User Guide

1. Dashboard Purpose & Overview

The Social Engine Revival dashboard is a research-grade analytical application built with Streamlit and Plotly. It provides pair-interactive visualization and live machine learning inference across the four stages of the Data Vortex competition.

2. Visual Design System

The interface adheres to a minimalist, professional analytical aesthetic: clean Inter typography, neutral slate/navy text, restrained indigo accenting, subtle borders, 10px rounded corners, and Plotly visualizations configured with crisp tooltips and no distracting rainbow palettes.

3. Information Architecture (4 Primary Sections)

The dashboard is strictly organized into four non-redundant sections accessible from the sidebar:

- **1. OVERVIEW:** Landing section with project title, subtitle, visual pipeline breadcrumb, 4 verified KPI cards (12,000 posts, 1,500 users, 9,000 training samples, 204 archive records), cross-round funnel chart, and concise inventory table.
- **2. SOCIAL INTELLIGENCE:** Reactive multi-filter panel (sentiment, entity, keyword, date range slider) with empty-state safety. Features four distinct tabs: Temporal Trends (sentiment index & monthly distribution), Entity & Term Frequency (gazetteer mentions & TF-IDF domain terms), Detected Shifts (candidate shifts SS1 & SS2), and Engagement / Signals (spikes ES1-ES3 plus compact SQLite query execution for E2, M1, H2).
- **3. NLP INTELLIGENCE:** Model architecture benchmark cards, training distribution charts, and a live post analysis engine supporting preset examples or custom text with calibrated probabilities and decision margins.
- **4. INSIGHTS:** Four structured synthesis cards (Data Integrity, Semantic Understanding, Social Signals, Engine Synthesis) and a prominent Methodology & Limitations callout.

4. Local Run Instructions

To launch the dashboard locally from the repository root:

```
python -m pip install -r Round-4/requirements.txt
streamlit run Round-4/app.py
```

The application defaults to port 8501 and loads all canonical assets relative to the repository root.